

MicroLED ADVANTAGES And HURDLES In Their Production

Despite its tangible benefits, microLED manufacturing is still too costly and full of challenges. But do the benefits outweigh the investments? If so, how soon can we expect them in displays?

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Apple, Samsung, TCL and all the leading tech giants in the market have been announcing new products that have microLED (μ LED) display screens. The industry is advancing with usage of μ LEDs in ultra-large as well as small-size displays. From television sets and computer monitors to mobile screens, tablets, virtual reality, automotive displays, and smart wearables, all are being taken over. With benefits ranging from incredible picture quality and efficient lighting control to saving power, this emerging tech is tempting mainstream industries and innovators alike.

“Players pursue μ LED displays because of their superior features. But that’s not the only major drive,” says Dr Xiaoxi He, Principal Analyst, IDTechEX, a market research firm. “The current LCD industry is mainly dominated by Mainland China. With South Korea taking the lead in organic light-emitting diodes (OLEDs), a new technology is being pushed to reshuffle the game in order to find new opportunities.”

While the term microLED refers to the size of the LEDs’ emitting area, there

is no clear industry standard about the size that qualifies as micro. However, it is being used to define LEDs that range from 100 μ m per side or less. Without any doubt, new techniques and methods to manufacture such devices would be required.

Despite the innate potential and pursuit of display innovators, this flat technology has not culminated in mass production yet. Let us take a look at the potential uses of this technology and the hurdles being faced in their commercial production.

μ LED benefits

Experts believe μ LEDs have good value propositions, such as colour gamut, high luminance, wide view angle, low power consumption, high dynamic range, high contrast, and transparency in comparison to other existing technologies. The ability to control each pixel for local dimming and brightening offers significant advantages.

Dr Xiaoxi explains that not all features are unique to μ LED, some can be noticed in alternative technologies as well. “Some features are great, but they may not be realised by current technology capability, or may not be so important for some applications, such as those that require low power consumption.”

When the LEDs become very small, their external quantum efficiency (EQE) also drops significantly. For lighting, normal sized LEDs can offer low power consumption, but tiny ones may not—unless technology improves further. In addition, for applications such as signage displays, the power consumption is a less significant issue compared to wearables and AR/



Fig. 1: Samsung Wall μ LED TV (Source: www.samsung.com)